MATH 163, Spring 2026
SCRIPT 15: Sequences

Definition 15.1. A *sequence* (of real numbers) is a function $a\colon \mathbb{N} \to \mathbb{R}$.

By setting $a_n = a(n)$, we can think of a sequence a as a list $a_1, a_2, a_3, \ldots$ of real numbers. We use the notation $(a_n)_{n=1}^{\infty}$ for such a sequence, or if there is no possibility of confusion, we sometimes abbreviate this and write simply (a_n). More generally, we also use the term sequence to refer to a function defined on $\{n \in \mathbb{N} \cup \{0\} \mid n \geq n_0\}$ for any fixed $n_0 \in \mathbb{N} \cup \{0\}$. We write $(a_n)_{n=n_0}^{\infty}$ for such a sequence.

Definition 15.2. We say that a sequence (a_n) *converges* to a point $p \in \mathbb{R}$ if, for every open interval I containing p, there exists $N \in \mathbb{N}$ such that if $n \geq N$, then $a_n \in I$. If a sequence converges to some point, we say it is *convergent*. If (a_n) does not converge to any point, we say that the sequence *diverges* or is *divergent*.

Exercise 15.3. Show that a sequence (a_n) converges to p if and only if any region containing p contains all but finitely many terms of the sequence, i.e., the set $\{n \in \mathbb{N} \mid a_n \notin R\}$ is finite.

Exercise 15.3. We wnat to show that a sequence (a_n) converges to a point $p \in \mathbb{R}$ if and only if any region containing p contains all but finitely many terms of the sequence.

($\Rightarrow$) Let a sequence (a_n) converge to $p \in \mathbb{R}$. Then, we have that for every $I \ni p$ there exists $N \in \mathbb{N}$ such that if $n \geq N$ then $a_n \in I$. As, by **Definition 15.2**, this applies for all I, let R be an arbitrary region containing p, and apply the definition with $I = R$.

As, by the definition of converging, we have that $\forall n \geq N$, $a_n \in R$. Therefore, there exists a set $[N-1]$ where $\forall x \in [N-1]$, $x \notin R$. Then, as $[N-1]$ is a finite set of natural numbers with N elements, we have that $\forall n \in \mathbb{N}$ where $a_n \notin R$ we have that $a_n \in [N]$. Therefore, the set $\{n \in \mathbb{N} \mid a_n \notin R\}$ is finite.

($\Leftarrow$) Let all regions R containing p containing all but finitely many terms of the sequence. Therefore, we have that $\{n \in \mathbb{N} \mid a_n \notin R\}$ is finite. Then, as $\{n \in \mathbb{N} \mid a_n \notin R\}$ is finite, we have that $\exists N \in \mathbb{N}$ such that if $n \geq N$ then $a_n \in R$. As this is true for every region R, we have that every open interval I containing p, there exists $N \in \mathbb{N}$ such that if $n \geq N$, then $a_n \in I$. This suffices to prove the statement.

This completes the proof. $\square$

Theorem 15.4. *Suppose that (a_n) converges both to p and to p'. Then $p = p'$.*

Proof of Theorem 15.4. Suppose that (a_n) converges to both p and to p'. Then, we want to show that $p = p'$.

Suppose for the sake of contradiction that that the sequence (a_n) converges to both p and to p' but $p \neq p'$. Since (a_n) converges to p, there exists $N \in \mathbb{N}$ such that if $n \geq N$ then $a_n \in (p-\delta, p+\delta)$. Since (a_n) converges to p', there exists $N' \in \mathbb{N}$ such that if $n \geq N'$ then $a_n \in (p'-\delta, p'+\delta)$.

Fix δ such that $\delta > 0$ and $p + \delta < p' - \delta$. Suppose, without loss of generality that $p < p'$. Then, we have that $\exists N \in \mathbb{N}$ such that if $n \geq N$ then $a_n \in (p-\delta, p+\delta)$. Now, consider the interval $(p'-\delta, p'+\delta)$. We have that if $n \geq N$ then $a_n \in (p-\delta, p+\delta)$. As this is true for all $n \geq N$, we have that $\nexists a_n \in (p'-\delta, p'+\delta)$. This contradicts the statement that the sequence (a_n) converges to both p and to p' where $p \neq p'$. Therefore, it must be that $p = p'$. □

Definition 15.5. If a sequence (a_n) converges to $p \in \mathbb{R}$, we call p the *limit* of (a_n) and write

$$\lim_{n\to\infty} a_n = p.$$

Exercise 15.6. Which of the following sequences (a_n) converge? Which diverge? For each that converges, what is the limit? Give proofs for your answers.

(a) $a_n = 5$. (b) $a_n = n$. (c) $a_n = \frac{1}{n}$. (d) $a_n = (-1)^n$.

Theorem 15.7. *A sequence (a_n) converges to $p \in \mathbb{R}$ if, and only if, for all $\epsilon > 0$, there is some $N \in \mathbb{N}$ such that for all $n \geq N$ we have $|a_n - p| < \epsilon$.*

Proof of Theorem 15.7. A sequence (a_n) converges to $p \in \mathbb{R}$ if, and only if, for all $\epsilon > 0$, there is some $N \in \mathbb{N}$ such that for all $n \geq N$ we have $|a_n - p| < \epsilon$

($\Rightarrow$) Let a_n converge to $p \in \mathbb{R}$. We want to show $\forall \epsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$ we have that $|a_n - p| < \epsilon$.

As the sequences a_n converges to $p \in \mathbb{R}$, we have that for every open interval I containing p, there exists $N \in \mathbb{N}$ such that if $n \geq N$ then $a_n \in I$. Then, we as have that this applies for every I, define $I = (p-\epsilon, p+\epsilon)$ for some $\epsilon > 0$. Then, as $a_n \in I$, we have that $p - \epsilon < a_n < p + \epsilon$. We can rewrite this inequality as $-\epsilon < a_n - p < \epsilon$. This is the equivalent of saying $|a_n - p| < \epsilon$.

As this applies for all intervals, we have that $\forall \epsilon > 9$ this works and there is some $N \in \mathbb{N}$ such that this works for all $n \geq N$ we have $|a_n - p| < \epsilon$.

($\Leftarrow$) Let $\epsilon > 0$ and suppose that there exists $N \in \mathbb{N}$ such that for all $n \geq N$ we have $|a_n - p| < \epsilon$. We want to show that (a_n) converges to p, i.e. that for every open interval I containing p, there exists $N \in \mathbb{N}$ such that if $n \geq N$ then $a_n \in I$.

Let I be an arbitrary open interval containing p. Then, we can write $I = (p - \epsilon, p + \epsilon)$ for some $\epsilon > 0$. As we have that $|a_n - p| < \epsilon$, we can rewrite this as $-\epsilon < a_n - p < \epsilon$, which gives us $p - \epsilon < a_n < p + \epsilon$. Therefore, $a_n \in I$. As this applies for every open interval I containing p, we have that (a_n) converges to p.

This completes the proof. $\square$

Exercise 15.8. (a) Prove that $\displaystyle\lim_{n\to\infty} \frac{(-1)^n}{n} = 0$.

(b) Let $x \in \mathbb{R}$ with $|x| < 1$. Prove that $\displaystyle\lim_{n\to\infty} x^n = 0$.

(c) Let $x \in \mathbb{R}$ with $|x| > 1$. Prove that (x^n) is divergent.

Hint for b) and c): Use Bernoulli's inequality (Sheet 0, Exercise 3) to show that if $|y| > 1$, then $|y|^n \geq n(|y| - 1) + 1$.

Theorem 15.9. *If $\displaystyle\lim_{n\to\infty} a_n = a$ and $\displaystyle\lim_{n\to\infty} b_n = b$ both exist, then*

(a) $\displaystyle\lim_{n\to\infty} (a_n + b_n) = a + b$;

(b) $\displaystyle\lim_{n\to\infty} (a_n \cdot b_n) = ab$.

(c) Moreover, if $b \neq 0$, then $\displaystyle\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{a}{b}$.

Exercise 15.10. Which of the following sequences (a_n) converge? Which diverge? For each that converges, what is the limit? Give proofs for your answers.

(a) $a_n = (-1)^n \cdot n$.

Proof. Suppose for sake of contradiction that $(-1)^n \cdot n$ converges. We know that $\frac{1}{n}$ converges, hence $(-1)^n \cdot n \cdot \frac{1}{n} = (-1)^n$ converges, which is a contradiction.

QED

(b) $a_n = \frac{1}{n^2+1}(2 + \frac{1}{n})$.

Proof. $\frac{1}{n^2+1} < \frac{1}{n} \longrightarrow_{n\to\infty} 0$, and $(2 + \frac{1}{n}) \longrightarrow_{n\to\infty} 2$. Hence, $a_n \longrightarrow_{n\to\infty} 0$.

QED

(c) $a_n = \frac{5n+1}{2n+3}$.

Proof. $\frac{5n+1}{2n+3} = \frac{5+\frac{1}{n}}{2+\frac{3}{n}} \longrightarrow_{n\to\infty} \frac{5+0}{2+0} = \frac{5}{2}$.

QED

(d) $a_n = \frac{(-1)^n+1}{n}$.

Proof. $\frac{(-1)^n+1}{n} = \frac{(-1)^n}{n} + \frac{1}{n} \longrightarrow_{n\to\infty} 0$.

QED

We've used the word "limit" in two contexts now: the limit points of a set, and the limit of a sequence. The definitions of these two terms may seem similar. Is there a formal connection? Theorem 15.11 alludes to an answer.

Theorem 15.11. *Let $A \subset \mathbb{R}$. Then $p \in \overline{A}$ if and only if there exists a sequence (a_n), with each $a_n \in A$, that converges to p.*

Definition 15.12. A sequence (a_n) is *bounded* if its image $\{a_n \mid n \in \mathbb{N}\}$ is bounded.

Theorem 15.13. *Every convergent sequence is bounded.*

Proof. Let $(a_n)_{n\in\mathbb{N}}$ be a sequence that converges to p. Let $(p-1, p+1)$ be a region containing p. Let $S = \{n \in \mathbb{N} \mid a_n \notin (p-1, p+1)\}$, then we know that S is finite. If $S = \emptyset$ then we are done.

If $S \neq \emptyset$ then let $m = \min\{a_n \mid a_n \notin (p-1, p+1)\}$ and $M = \max\{a_n \mid a_n \notin (p-1, p+1)\}$. We know that the minimum and maximum exist since $\{a_n \mid a_n \notin (p-1, p+1)\}$ is finite.

Then, $\min(m, p-1)$ is a lower bound for $(a_n)_{n\in\mathbb{N}}$ and $\max(M, p+1)$ is an upper bound for $(a_n)_{n\in\mathbb{N}}$.

This completes the proof.

QED

Exercise 15.14. Construct a sequence with two subsequential limits. Construct a sequence with infinitely many subsequential limits.

Proof. Let $a_n = (-1)^n$. Note that a_n does not converge, but we can extract subsequences that do. Let $i : \mathbb{N} \to \mathbb{N}$ be defined by $i(x) = 2x$. Then, $b_n = a_{i(n)} = a_{2n} = 1$ for all n, hence b_n converges to 1, giving a subsequential limit of 1. Let $j : \mathbb{N} \to \mathbb{N}$ be defined by $j(x) = 2x+1$. Then, $c_n = a_{j(n)} = a_{2n+1} = -1$ for all n, hence c_n converges to -1, giving a subsequential limit of -1. Thus, a_n has two subsequential limits.
Let m_n be a sequence defined as follows.

$$m_n = \begin{cases} 1, & \text{if } n = 1 \\ p, & \text{if } n \neq 1 \text{ and } p \text{ is the least prime divisor of } n. \end{cases}$$

We know that there are infinitely many prime numbers. For every prime p, the subsequence $m_p, m_{p^2}, m_{p^3}, \ldots = p, p, p, \ldots$ is constant and hence converges to p, giving p as a subsequential limit. Since there are infinitely many primes, we obtain infinitely many distinct subsequential limits.
This completes the proof. QED

Proof of 15.11 that waS done in class. Let $A \subset \mathbb{R}$. such taht $p \in \overline{A}$. iff $\exists a_n$ such that a = lim x -¿ a an = p (comverges to p)

Let $p \in \overline{A}$.

$p \in A.15.6(a)a_n : \mathbb{N} \to \mathbb{R}$

$p \in a$

$\square$

Theorem 15.15. *Every bounded sequence has a convergent subsequence.*

(Hint: Use Theorem 10.21.)

We are now able to prove a useful characterization of convergent sequences.

Theorem 15.16. *A sequence* (a_n) *of real numbers converges if, and only if, for all* $\epsilon > 0$, *there is some* $N \in \mathbb{N}$ *such that* $|a_n - a_m| < \epsilon, \forall n \geq N, m \geq N$.

Exercise 15.17. Use Theorem 15.16 to show that the sequence in Exercise 15.10a) does not converge.

Additional Exercises

1. i) Suppose that for two convergent sequences of real numbers, (a_n) and (b_n), $a_n \leq b_n$, for all $n \in \mathbb{N}$. Prove that $\lim_{n\to\infty} a_n \leq \lim_{n\to\infty} b_n$.

   ii) Suppose that $0 \leq a_n \leq b_n$, for all $n \in \mathbb{N}$. Suppose $\lim_{n\to\infty} b_n = 0$. Show that $\lim_{n\to\infty} a_n = 0$ also.

   iii) Suppose that $(a_n), (b_n), (c_n)$ are sequences of real numbers such that $a_n \leq b_n \leq c_n$, for all $n \in \mathbb{N}$. Suppose also that $\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = L$. Prove that $\lim_{n\to\infty} b_n = L$ also.

   iv) Let $a \in \mathbb{R}$ with $a > 1$. Prove that $\lim\limits_{n\to\infty} a^{\frac{1}{n}} = 1$. Does this result hold for $0 < a \leq 1$ also? *Hint: Use iii) and Bernoulli's inequality (Script 0, Exercise 3) with an appropriate choice of* x.

   v) Justify the following inequalities:

   $$\left(1 + \frac{1}{\sqrt{n}}\right)^n \geq 1 + \sqrt{n} \geq \sqrt{n}.$$

   Hence prove that $\lim_{n\to\infty} n^{\frac{1}{n}} = 1$.

2. Suppose that the sequence (a_n) converges to 0 and the sequence (b_n) is bounded. Prove that

   $$\lim_{n\to\infty} (a_n b_n) = 0.$$

3. Suppose that (a_n) is a sequence of real numbers. Prove that (a_n) has either an increasing subsequence or a decreasing subsequence (or both).

4. For $n \in \mathbb{N}$, define $a_{n+1} = 2 + a_n/3$ and let $a_0 = 100$. Show that (a_n) is convergent.

5. Find the limit of each of the following sequences:

(a) $a_n = \sqrt{n+1} - \sqrt{n-1}$;

(b) $a_n = n - \sqrt{n+1}\sqrt{n+2}$.

6. Let $a > 0$ and consider a sequence

$$x_n = \sqrt{a + \sqrt{a + ...\sqrt{a}}},$$

where n is the number of roots. Prove that x_n converges and find its limit. You may assume that square roots are well-defined, i.e. there is a bijective, continuous, strictly increasing function $g : \mathbb{R}^+ \to \mathbb{R}^+$ such that $g(x)^2 = x$ for all $x \in \mathbb{R}^+$.

Here $\mathbb{R}^+$ denotes the set of non-negative real numbers.

7. Let $f : [a, b] \to \mathbb{R}$ be a bounded function and $\Omega \in \mathbb{R}$. Prove that $\int_a^b f$ exists and equals Ω if and only if there exists a sequence of partitions $(P_n)_{n\in\mathbb{N}}$ such that

$$\lim_{n\to\infty} U(P_n, f) = \lim_{n\to\infty} L(P_n, f) = \Omega.$$

8. a) Let $f : [a, b] \longrightarrow \mathbb{R}$ be a bounded function. Let P_n be a partition of $[a, b]$ into n equal subintervals. Prove that if $\lim_{n\longrightarrow\infty} U(f, P_n) = \lim_{n\longrightarrow\infty} L(f, P_n) = L$, then f is integrable on $[a, b]$ and $\int_a^b f = L$.

   b) Use a) to prove that $f(x) = x^2$ is integrable on $[a, b]$ and find $\int_a^b f$. *Hint: Sheet 0 Exercise 2 should be useful.*

9. Let A be an open set and $f\colon A \to \mathbb{R}$.

   a) Write down the negation of the statement: "Given $\epsilon > 0$, there is some $\delta > 0$ such that if $x \in A$ and $|x - a| < \delta$, then $|f(x) - f(a)| < \epsilon$."

   b) Prove that if $f\colon A \longrightarrow \mathbb{R}$ is not continuous at $a \in A$, then there is some $\epsilon > 0$ and a sequence (x_n) in A such that $|x_n - a| < \frac{1}{n}$ and $|f(x_n) - f(a)| \geq \epsilon$.

   c) Prove that a function $f\colon A \longrightarrow \mathbb{R}$ is continuous at $a \in A$ if, and only if, $\lim_{n\longrightarrow\infty} f(x_n) = f(a)$ for every sequence (x_n) in A such that $\lim_{n\longrightarrow\infty} x_n = a$.

   d) Prove that a function $f\colon A \longrightarrow \mathbb{R}$ is uniformly continuous if, and only if, for every pair of sequences $(x_n), (y_n)$ in A such that $(x_n - y_n)$ converges to 0, $(f(x_n) - f(y_n))$ converges to 0.

10. We say that a sequence is *a close sequence* if it satisfies the condition in Theorem 15.16, i.e. if, for all $\epsilon > 0$, there is some $N \in \mathbb{N}$ such that $|a_n - a_m| < \epsilon$, for all $n \geq N, m \geq N$. Suppose that $f\colon A \longrightarrow \mathbb{R}$ is uniformly continuous and (x_n) is a close sequence in A. Prove that $(f(x_n))$ is a close sequence in $\mathbb{R}$.

11. Suppose that $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ is a continuous function.

   (a) Show that if f is uniformly continuous, then $\lim_{x\to 0} f(x)$ exists.

   (b) Give an example to show that $\lim_{x\to 0} f(x)$ need not exist if f is only continuous (but not uniformly).

12. Suppose that $A \subset B$ and that A is dense in B. Suppose $f : A \to \mathbb{R}$ is uniformly continuous. Prove that there exists a unique continuous function $\overline{f} : B \to \mathbb{R}$ such that $f(x) = \overline{f}(x)$ for all $x \in A$. Moreover, $\overline{f}$ is uniformly continuous.

13. (a) A *close sequence* (of rational numbers) is a sequence (a_n) with $a_n \in \mathbb{Q}$ such that, given an $\epsilon \in \mathbb{Q}$ with $\epsilon > 0$, there is some $N \in \mathbb{N}$ such that for all $n, m \geq N$ we have $|a_n - a_m| < \epsilon$. (This is the condition we see in Theorem 15.16.) Prove that

    i) Any sequence of rationals that converges in $\mathbb{Q}$ is a close sequence.

    ii) Close sequences are bounded.

    iii) If (x_n) and (y_n) are close sequences then so are $(x_n + y_n)$ and $(x_n y_n)$.

    iv) The set of close sequences of rationals has additive and multiplicative identities, additive inverses, but many close sequences fail to have multiplicative inverses.

    (b) Define X to be the set of all close sequences of rational numbers. Then we define an equivalence relation on X by $(a_n) \sim (b_n)$ if $\lim\limits_{n \longrightarrow \infty} |a_n - b_n| = 0$. Prove that this is indeed an equivalence relation.

    We let $\mathcal{R}$ denote the set of equivalences classes of close sequences of rationals and define addition and multiplication in $\mathcal{R}$ by

$$\begin{aligned} [(a_n)] + [(b_n)] &= [(a_n + b_n)] \\ [(a_n)][(b_n)] &= [(a_n b_n)]. \end{aligned}$$

    Note that we can view $\mathbb{Q}$ as a subset of $\mathcal{R}$ by identifying a rational number q with the equivalence class of the constant sequence, $[(q, q, q, ...)]$.

    (c) Prove that addition and multiplication are well-defined.

    (d) Prove that $\mathcal{R}$ is a field.

    Note: The field axioms should all be very routine, except for FA8.

    (e) We say that an equivalence class $[(a_n)]$ is *positive* and write $[(a_n)] > 0$, or $0 < [(a_n)]$, if there is some $N \in \mathbb{N}$ and $\epsilon \in \mathbb{Q}, \epsilon > 0$, such that $a_n > \epsilon$ for all $n \geq N$. We say that $[(a_n)] > [(b_n)]$ (or $[(b_n)] < [(a_n)]$), if $[(a_n - b_n)] > 0$. Prove that $<$ is well-defined and that $\mathcal{R}$ is an ordered field with this ordering.

    (f) i) Prove that $\mathcal{R}$ satisfies Axioms 1-3.

    ii) Prove that if $x, y \in \mathcal{R}, x < y$, there is some $z \in \mathcal{R}$ such that $x < z < y$.

    iii) Prove every nonempty bounded subset of $\mathcal{R}$ has a supremum.

    iv) Does $\mathcal{R}$ satisfy Axiom 4?

    v) Does $\mathcal{R}$ satisfy Axiom 5?

    Hint for iii): Let X be a bounded subset of $\mathcal{R}$. Consider

$$Y = \{q \in \mathbb{Q} \mid [(q, q, q, \cdots)] \textit{ is not an upper bound of } X\}.$$

    Construct a suitable sequence (q_n) of points in Y.

    (g) What can you say about $\mathcal{R}$?